

The direction field associated to a planar diff. sys

$$(1.) \quad \dot{x} = f(x)$$

$$f \in C^1(\mathbb{R}^2, \mathbb{R}^2)$$

$$X = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$f = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix} \quad (1.) \quad \begin{cases} \dot{x} = f_1(x, y) \\ \dot{y} = f_2(x, y) \end{cases}$$

Recall: a second order eq. $\ddot{x} = f(\dot{x}, x)$ can be written as a planar system with the unknown $X = \begin{pmatrix} x \\ \dot{x} \end{pmatrix}$ i.e. $y = \dot{x}$

$$\ddot{x} = f(\dot{x}, x) \Leftrightarrow \begin{cases} \dot{x} = y \\ \dot{y} = f(y, x) \end{cases}$$

The notion of first integral

$$(x, y) \in U \subset \mathbb{R}^2 \mapsto H(x, y) \in \mathbb{R}.$$

H is a first integral of (1) when H is not locally constant, but H is constant on each orbit.

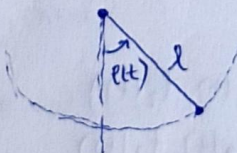
$t \mapsto H(\varphi(t))$ is a constant function (does not depend on t) for each solution φ of (1)

first integral (math) $\rightarrow$ conserved quantity (from physics)
(for example, the total energy)

Ex: We consider the non-linear simple pendulum equation:

$$\ddot{x} + \omega^2 \sin x = 0, \quad (\omega > 0 \text{ fixed positive parameter})$$

Let $\varphi: \mathbb{R} \rightarrow \mathbb{R}^2$ be a solution, i.e. $\varphi \in C^2(\mathbb{R})$ and $\varphi''(t) + \omega^2 \sin \varphi(t) = 0, (\forall) t \in \mathbb{R}.$



$$E_p(t) = \underbrace{\frac{1}{2} [\dot{\varphi}(t)]^2}_{\text{kinetic energy}} + \underbrace{\omega^2 \cdot (1 - \cos \varphi(t))}_{\text{potential energy}}, \quad (\forall) t \in \mathbb{R}.$$

(fct. is c.t.)

We prove that the energy is conserved, i.e. $E_p(t) = E_p(0)$, $(\forall) t \in \mathbb{R}$.

$$\begin{aligned} \frac{d}{dt} E_p(t) &= \frac{1}{2} \cdot 2 \cdot \dot{\varphi}(t) \cdot \ddot{\varphi}(t) + \omega^2 \cdot \sin \varphi(t) \cdot \dot{\varphi}(t) \\ &= \dot{\varphi}(t) \cdot \underbrace{(\ddot{\varphi}(t) + \omega^2 \cdot \sin \varphi(t))}_0 = 0, \quad (\forall) t \in \mathbb{R}. \end{aligned}$$

Now, using what we proved, find a first integral of ...

$$\begin{cases} \dot{x} = y \\ \dot{y} = -\omega^2 \cdot \sin x \end{cases}$$

$x = \varphi(t)$
 $y = \dot{\varphi}(t)$

A solution $\varphi: \mathbb{R} \rightarrow \mathbb{R}^2$ of the system has the components $\varphi = \begin{pmatrix} \varphi \\ \dot{\varphi} \end{pmatrix}$ where φ is a sol. of $\ddot{x} + \omega^2 \sin x = 0$.

$$H: \mathbb{R}^2 \rightarrow \mathbb{R}, \quad H(x, y) = \frac{1}{2} y^2 + \omega^2 \cdot (1 - \cos x)$$

Theorem (How to check that a given H is a first integral)

Let $U \subset \mathbb{R}^2$ domain, $H: U \rightarrow \mathbb{R}$ be a function, not locally constant.

We have the following equivalence:

$$\boxed{f_1(x, y) \frac{\partial H}{\partial x}(x, y) + f_2(x, y) \frac{\partial H}{\partial y}(x, y) = 0, \quad (\forall) (x, y) \in U} \quad \text{PDE}$$

$\Leftrightarrow H$ is a first integral of U of (1)

Proof.

only " $\Rightarrow$ "

Let φ be a solution of (1).

the derivative should be constant

$$\begin{aligned} \frac{d}{dt} H(\varphi(t)) &= \frac{d}{dt} H(\varphi_1(t), \varphi_2(t)) \quad \underline{\text{Chain rule}} \\ &= \frac{\partial H}{\partial x}(\varphi_1(t), \varphi_2(t)) \cdot \dot{\varphi}_1(t) + \frac{\partial H}{\partial y}(\varphi_1(t), \varphi_2(t)) \cdot \dot{\varphi}_2(t) \end{aligned}$$

$$\varphi = \begin{pmatrix} \varphi_1 \\ \varphi_2 \end{pmatrix} \Rightarrow \begin{cases} \dot{\varphi}_1(t) = f_1(\varphi_1(t), \varphi_2(t)) \\ \dot{\varphi}_2(t) = f_2(\varphi_1(t), \varphi_2(t)) \end{cases}$$

$$= \frac{\partial H}{\partial x}(\varphi(t)) \cdot \dot{\varphi}_1(t) + \frac{\partial H}{\partial y}(\varphi(t)) \cdot \dot{\varphi}_2(t) \stackrel{\text{hyp}}{=} 0.$$

$\Rightarrow H$ first integral.

Ex: ① Check that $H: \mathbb{R}^2 \rightarrow \mathbb{R}$, $H(x, y) = x^2 + y^2$ is a first integral of $\begin{cases} \dot{x} = -y \\ \dot{y} = x \end{cases}$

$\rightarrow H \in C^1(\mathbb{R}^2)$

$\nabla H = (2x, 2y)$, H not locally constant

$\rightarrow$ We have to check: $\underbrace{-y}_{2x} \cdot \underbrace{\frac{\partial H}{\partial x}}_{2x} + \underbrace{x}_{2y} \cdot \underbrace{\frac{\partial H}{\partial y}}_{2y} = 0$, $(\forall) (x, y) \in \mathbb{R}^2$ TRUE.
Theorem, H is a f.i.

A procedure to find the first integral of (1):

$\rightarrow$ Step 1: $\frac{dy}{dx} = \frac{f_2(x, y)}{f_1(x, y)}$

$\rightarrow$ Step 2: find the gen. sol. and write it in the form:

$$H(x, y) = c, \quad c \in \mathbb{R}$$

$\rightarrow$ Step 3: take $H: U \rightarrow \mathbb{R}$ and check whether it is a first integral.

How to find the general solution of a separable eq:

$$\frac{dy}{dx} = g_1(x) \cdot g_2(y)$$

$\rightarrow$ Step 1: separate the variables:

$$\frac{dy}{g_2(y)} = g_1(x) dx$$

$\rightarrow$ Step 2: integrate

$$G_2(y) = \int \frac{dy}{g_2(y)}$$

$$G_1(x) = \int g_1(x) dx$$

and write the sol. as: $G_2(y) = G_1(x) + C$, $C \in \mathbb{R}$.

If possible, simplify this relation

Ex. $\begin{cases} \dot{x} = -y \\ \dot{y} = x \end{cases}$ Find a first integral:

→ Step 1: $\frac{dy}{dx} = \frac{x}{-y}$

→ Step 2: the equation is separable:

$$-y dy = x dx \quad | \int$$

$$\Rightarrow -\frac{y^2}{2} = \frac{x^2}{2} + C$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} = C, \quad C \in \mathbb{R}. \quad | \cdot 2$$

$$\Rightarrow \underline{x^2 + y^2 = C.}$$

→ Step 3: $H(x, y) = x^2 + y^2$, $(\forall) (x, y) \in \mathbb{R}^2$, $H: \mathbb{R}^2 \rightarrow \mathbb{R}$.

Problem: Set $\omega > 0$ and $\ddot{x} + \omega^2 \cdot \sin x = 0$

(i) Find a first integral of this planar system.

(ii) Justify that the equilibrium position $\varphi^*(t) = 0$, $t \in \mathbb{R}$ is stable.

→ Second-order ^{non}linear diff. equation

→ Equivalent planar system: $\begin{cases} \dot{x} = y & \text{(always)} \\ \dot{y} = -\omega^2 \cdot \sin x & \text{only } x, y \end{cases}$
↑ only derivatives.

→ Step 1: $\frac{dy}{dx} = \frac{-\omega^2 \cdot \sin x}{y}$ (separable).

$$\Rightarrow y dy = -\omega^2 \cdot \sin x dx \quad | \int$$

$$\Rightarrow \frac{y^2}{2} = +\omega^2 (\cos x + C)$$

$$\Rightarrow y^2 = +2\omega^2 \cdot \cos x + C$$

$$\Rightarrow \underbrace{y^2 + 2\omega^2 \cdot \cos x}_{H(x, y)} = C, \quad C \in \mathbb{R}$$

→ Step 2: $H(x, y) = y^2 + 2\omega^2 \cdot \cos x$, $(\forall) (x, y) \in \mathbb{R}^2$, $H \in C^1(\mathbb{R}^2)$

→ Step 3: Check: $\frac{\partial H}{\partial x} \cdot y + (-\omega^2 \cdot \sin x) \cdot \frac{\partial H}{\partial y} \stackrel{?}{=} 0$, $(\forall) (x, y) \in \mathbb{R}^2$?

(4)

$$\Rightarrow y \cdot (2w^2 \cdot \sin x) - w^2 \cdot \sin x \cdot 2y \stackrel{?}{=} 0 \text{ true.}$$

$\Rightarrow H: \mathbb{R}^2 \rightarrow \mathbb{R}, H(x, y) = y^2 - 2w^2 \cos x$ is a first integral.

ii)



equilibrium position

stable $\rightarrow$ it won't move too much from equilibrium

$\dot{x}(t) = 0, t \in \mathbb{R}$ is a solution of $\ddot{x} + w^2 \sin x = 0$

it corresponds to $(0, 0) \rightarrow$ equilibrium point of $\begin{cases} \dot{x} = y \\ \dot{y} = -w^2 \sin x \end{cases}$

So, we have to study the stability of the equil. point $(0, 0)$ of this non-linear system.

Linearization method:

$$f(x, y) = \begin{pmatrix} y \\ -w^2 \sin x \end{pmatrix}$$

$$Jf(x, y) = \begin{pmatrix} 0 & 1 \\ -w^2 \cos x & 0 \end{pmatrix}$$

$$A = Jf(0, 0) = \begin{pmatrix} 0 & 1 \\ -w^2 & 0 \end{pmatrix}$$

eigenvalues: $\det(A - \lambda I_2) = 0$

$$\Rightarrow \begin{vmatrix} -\lambda & 1 \\ -w^2 & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 + w^2 = 0$$

$$\Rightarrow \lambda = \pm iw$$

Then the linear system $\dot{x} = AX$ is a center, stable.

Is the equilibrium $(0, 0)$ hyperbolic? No, because $\operatorname{Re}(\lambda_1) = \operatorname{Re}(\lambda_2) = 0$

$\Rightarrow$ equil. is non-hyperbolic

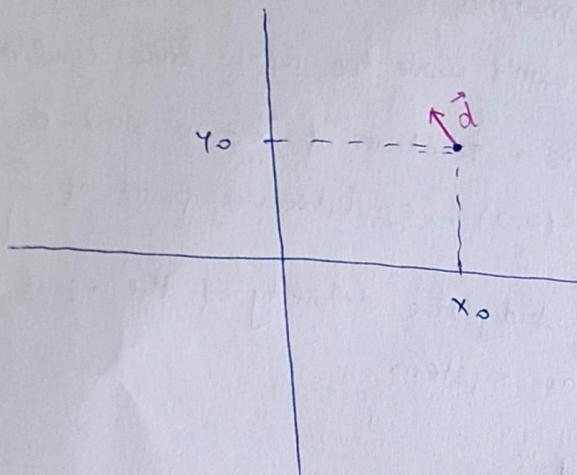
$\Rightarrow$ Lin. method fails.

We have: the eigenvalues of the linearization around $(0, 0)$ ^{are} $\lambda_{1,2} = \pm iw$ and $\exists H: \mathbb{R}^2 \rightarrow \mathbb{R}$ a first integral. Theorem $\rightarrow (0, 0)$ stable.

2) The direction field.

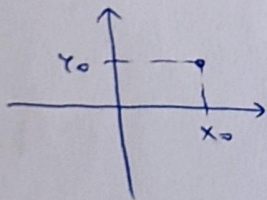
$$(1) \begin{cases} \dot{x} = f_1(x, y) \\ \dot{y} = f_2(x, y) \end{cases}$$

In each point $(x_0, y_0) \in \mathbb{R}^2$ we have a vector $\vec{d}$ of coordinates $\vec{d}(f_1(x_0, y_0), f_2(x_0, y_0))$

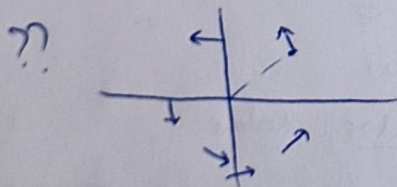


$$\text{Ex: } \begin{cases} \dot{x} = -y \\ \dot{y} = x \end{cases}$$

In each point (x_0, y_0) , the direction field is orthogonal to the vector of coords. (x_0, y_0)



$\vec{d}(x_0, y_0)$ has the coordinates $(-y_0, x_0)$.
The vectors (x_0, y_0) , $(-y_0, x_0)$ are orthogonal since their scalar prod = 0.



Let $\gamma = \begin{pmatrix} \gamma_1 \\ \gamma_2 \end{pmatrix}$ be a sol. of (1)

The corresponding orbit is the planar curve γ of parametric equations:

$$\begin{cases} x = \gamma_1(t) \\ y = \gamma_2(t) \end{cases}$$

It is known that the vector $(\dot{\gamma}_1(t_0), \dot{\gamma}_2(t_0))$ is tangent to γ in the point $(x_0, y_0) = (\gamma_1(t_0), \gamma_2(t_0))$

$$\dot{x}_1(t_0) = f_1(\ell(t_0)) = f_1(x_0, y_0)$$

$$\dot{x}_2(t_0) = f_2(\ell(t_0)) = f_2(x_0, y_0)$$

direction + vectoriel de position in
orbit point

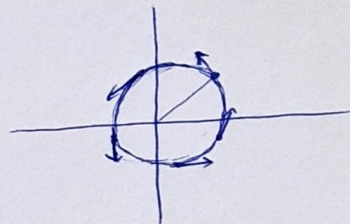
Then, the direction $\vec{d}$ in (x_0, y_0) of coord $(f_1(x_0, y_0), f_2(x_0, y_0))$ is tangent to the orbit of the system.

↳ Sol. of curve?

$$\text{Ex: } \begin{cases} \dot{x} = -y \\ \dot{y} = x \end{cases}$$

What is the shape of the orbits?

⇒ circles.



$$y' = g(x, y)$$

The direction field:

$\vec{d}$, based in $(x_0, y_0) \in \mathbb{R}^2$ has the shape $g(x_0, y_0)$

circle → tg ⊥ radius

We have that $\vec{d}$ is tangent to the graph of a solution.

$$\text{Remark: } \begin{cases} \dot{x} = f_1(x, y) \\ \dot{y} = f_2(x, y) \end{cases}$$

have the same direction field.
(orbits as graphs of solutions)

$$\frac{dy}{dx} = \frac{f_2(x, y)}{f_1(x, y)}$$

same
(slope)

$$\text{Ex: } y' = 1 - \frac{x}{y^2} \text{ Dir. field in } (1, 1), (1, 0), (0, 1).$$

$$\text{Define } g(x, y) = 1 - \frac{x}{y^2}$$

$$g(1, 1) = 1 - 1 = 0$$

$$g(1, 0) = 1 - \frac{1}{0} = -\infty$$

$$g(0, 1) = 1 - 0 = 1$$

